

Snort - Intrusion Detection System (IDS)

Introduction to Snort

What is Snort?

Snort is an Open-Source Network Intrusion Detection System (NIDS) used to monitor network traffic and detect suspicious or malicious activities based on predefined rules.

Snort can perform:

- Real-time traffic analysis
- Packet logging
- Intrusion detection
- Protocol analysis
- Network monitoring
- Security alert generation

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Intrusion Detection System (IDS)

An Intrusion Detection System (IDS) monitors network traffic and identifies suspicious activities, attacks, or policy violations.

Purpose of IDS

- Detect unauthorized access
 - Detect malware activity
 - Detect network scans
 - Detect suspicious traffic
 - Generate alerts
-

IDS vs IPS

IDS	IPS
Detects attacks	Detects and blocks attacks

IDS	IPS
Passive monitoring	Active protection
Generates alerts	Takes preventive actions
Does not block traffic	Can block malicious traffic

Types of IDS

1. Network IDS (NIDS)

Monitors network traffic.

Examples:

- Snort
- Suricata
- Zeek

2. Host IDS (HIDS)

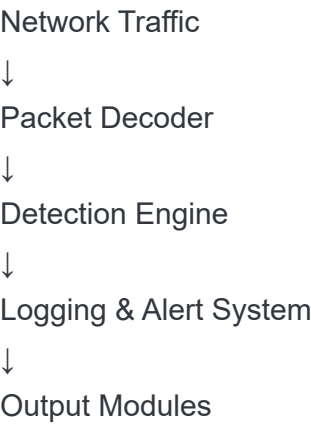
Monitors activity on a specific system.

Examples:

- OSSEC
- Wazuh

Snort Architecture

Snort processes traffic through several components.



Packet Decoder

Responsible for decoding network packets.

Protocols:

- Ethernet
 - IP
 - TCP
 - UDP
 - ICMP
-

Detection Engine

The core component of Snort.

Responsibilities:

- Match traffic against rules
 - Detect attacks
 - Generate alerts
-

Logging and Alerting

Responsible for:

- Generating alerts
 - Storing logs
 - Reporting suspicious activities
-

Snort Installation

Update Repository

```
sudo apt update
```

Install Snort

```
sudo apt install snort -y
```

Verify Installation

```
snort -V
```

Expected Output:

Identify Network Interface

```
ip a
```

Example Output:

```
ens33  
eth0  
ens160
```

Use your active network interface.

Important Snort Files

Main Configuration File

```
/etc/snort/snort.conf
```

Purpose:

- Network variables
 - Rule paths
 - Alert settings
 - Detection settings
-

Local Rules File

```
/etc/snort/rules/local.rules
```

Purpose:

Store custom rules created by administrators.

Log Directory

```
/var/log/snort
```

Purpose:

Stores alerts and logs.

Understanding snort.conf

Open file:

```
sudo nano /etc/snort/snort.conf
```

HOME_NET Variable

Example:

```
var HOME_NET 192.168.1.0/24
```

Represents the internal network to be monitored.

EXTERNAL_NET Variable

Example:

```
var EXTERNAL_NET any
```

Represents external networks.

Testing Configuration

Always test configuration before running Snort.

```
sudo snort -T -c /etc/snort/snort.conf
```

Expected Output:

```
Snort successfully validated the configuration
```

Snort Rule Basics

Rules tell Snort what activity should be detected.

Rule Syntax

```
action protocol source_ip source_port -> destination_ip destination_port (options)
```

Example:

```
alert icmp any any -> any any (msg:"Ping Detected"; sid:1000001; rev:1;)
```

Rule Components

Action

Defines what Snort should do.

Common actions:

```
alert
log
pass
drop
reject
```

Protocol

Defines traffic type.

Examples:

```
tcp
udp
icmp
ip
```

Source Address

```
any
```

Means traffic from any source.

Destination Address

```
any
```

Means traffic to any destination.

Rule Options

Options are written inside brackets.

Example:

```
(msg:"Ping Alert"; sid:1000001; rev:1;)
```

msg

Alert message.

Example:

```
msg:"ICMP Ping Detected"
```

sid (Signature ID)

Unique rule identifier.

Example:

```
sid:1000001
```

Rules should always have unique SID values.

gid (Generator ID)

Identifies the module generating the alert.

Default:

```
gid:1
```

rev (Revision)

Rule version number.

Example:

```
rev:1
```

Creating First Snort Rule

Open local rules file:

```
sudo nano /etc/snort/rules/local.rules
```

Add:

```
alert icmp any any -> any any (msg:"Ping Detected"; sid:1000001; rev:1;)
```

Save file.

Validate Rule

```
sudo snort -T -c /etc/snort/snort.conf
```

Start Snort

```
sudo snort -A console -q -c /etc/snort/snort.conf -i ens33
```

Generate Traffic

From another machine:

```
ping target-ip
```

Expected:

```
Ping Detected
```

Alert appears.

Custom Rule Examples

SSH Detection

```
alert tcp any any -> any 22 (msg:"SSH Connection Detected"; sid:1000002; rev:1;)
```

HTTP Detection

```
alert tcp any any -> any 80 (msg:"HTTP Traffic Detected"; sid:1000003; rev:1;)
```

HTTPS Detection

```
alert tcp any any -> any 443 (msg:"HTTPS Traffic Detected"; sid:1000004; rev:1;)
```

Telnet Detection

```
alert tcp any any -> any 23 (msg:"Telnet Access Detected"; sid:1000005; rev:1;)
```

FTP Detection

```
alert tcp any any -> any 21 (msg:"FTP Access Detected"; sid:1000006; rev:1;)
```

Detecting Port Scans

SYN Scan Rule

```
alert tcp any any -> any any (flags:S; msg:"Possible SYN Scan"; sid:1000007; rev:1;)
```

Practical Testing

Test HTTP Rule

```
curl http://example.com
```

Test HTTPS Rule

```
curl https://example.com
```

Test SSH Rule

```
ssh user@server-ip
```

Test Nmap Scan

```
nmap target-ip
```

Alert Logging

View alerts:

```
sudo tail -f /var/log/snort/alert
```

Common Snort Commands

Check Version

```
snort -V
```

Check Configuration

```
sudo snort -T -c /etc/snort/snort.conf
```

Sniffer Mode

```
sudo snort -v
```

IDS Mode

```
sudo snort -A console -q -c /etc/snort/snort.conf -i ens33
```

Packet Logging Mode

```
sudo snort -dev -l ./logs
```

Real-Time Lab Demonstration

Machine 1 (Victim/Monitoring System)

```
sudo snort -A console -q -c /etc/snort/snort.conf -i ens33
```

Machine 2 (Attacker/Test Machine)

```
ping victim-ip
```

```
curl http://victim-ip
```

```
ssh user@victim-ip
```

```
nmap victim-ip
```

Students will observe:

- ICMP Detection
 - HTTP Detection
 - SSH Detection
 - Port Scan Detection
 - Real-Time Alert Generation
-

Assignment

Task 1

Install Snort and verify version.

Task 2

Configure HOME_NET in snort.conf.

Task 3

Create ICMP Detection Rule.

Task 4

Create SSH Detection Rule.

Task 5

Create HTTP Detection Rule.

Task 6

Create HTTPS Detection Rule.

Task 7

Perform an Nmap scan and capture the generated alert.

Task 8

Document all alerts observed during testing.

Conclusion

- ✓ Install Snort
- ✓ Configure Snort
- ✓ Understand Snort Architecture
- ✓ Write Custom Rules
- ✓ Generate Alerts
- ✓ Monitor Network Traffic
- ✓ Detect Basic Network Attacks
- ✓ Perform Basic IDS Monitoring